ANALYSIS SUMMARY

OF

WISDOM NET: TRUSTABLE MACHINE LEARNING TOWARD ERROR-FREE CLASSIFICATION

PROBLEM AND MOTIVATION:

Even classifiers with over 95% accuracy can make incorrect predictions. In critical fields such as medical diagnosis, disaster prediction, fraud detection, and protein crystallization, these mistakes can be costly or even dangerous. Manually reviewing predictions reduces the benefit of automation. WisdomNet addresses this through "trustable learning," where the model learns to recognize cases it is likely to misclassify and abstains (rejects) instead of making an unreliable prediction. Unlike approaches that require manually labeling a separate "likely-lead" class, WisdomNet identifies uncertain cases automatically from the base model's own misclassifications.

THE WISDOM NET ARCHITECTURE:

WisdomNet can be added to an already-trained neural network, such as an MLP, CNN, or ResNet. It works in four steps:

1. **Pre-train**: Train a base network **N** on the complete training set **S** to perform the classification task.
2. **Identify Misclassifications**: Run **N** on **S** and collect all incorrectly classified samples into a reject set **(S_r)**.
3. **Add a Reject Neuron**: Copy **N** into a new network **W** and add one output neuron called the **conjugate neuron**, representing the **reject** class. Its weights and bias initially start at zero, so W behaves exactly like N.
4. **Fine-tune on (S_r)**: Train **W** only on the reject set S_r. Since this set is small, the network can learn these difficult cases without having to retrain the entire classification task.

At inference, **W** can produce either a class prediction or a rejection. During fine-tuning, some samples that N previously classified correctly may also be rejected. These are considered "lucky" predictions, cases where the model got the answer right without reliably learning the underlying pattern. By identifying both known mistakes and potentially unreliable predictions, WisdomNet makes automated classification more trustworthy.

THEORETICAL SUPPORT FOR THE ARCHITECTURE:

Two theorems support the WisdomNet architectural design.

- **Theorem 1 (Equivalence):** When the zero-initialized conjugate neuron is added, **W** behaves exactly like the original network **N**. This is because the new neuron initially contributes nothing, while all existing weights remain unchanged.
- **Theorem 2 (Convergence):** The fine-tuning process can achieve zero training error on the reject set $\mathbf{S}_r$. Since S_r is much smaller than the full training set, the network has more than enough parameters to learn these difficult cases. Based on prior work by Soudry & Carmon (2016), the authors argue that the local minima in this setting are also global minima, allowing the network to fit the reject samples without error.

EXPERIMENTAL RESULTS:

WisdomNet was tested on distinct models, including MLPs on the Breast Cancer, Banknote, and Ionosphere datasets, a CNN on MNIST 4-vs-9, and a ResNet50 on Dogs-vs-Cats. In all experiments, WisdomNet reduced the classification error to 0% while rejecting only a small percentage of samples.

Dataset / Model	Base Error	WisdomNet Error	Reject Rate
BC (MLP)	1.76%	0%	10.05%
BN (MLP)	1.00%	0%	1.50%
IO (MLP)	3.71%	0%	13.58%
MNIST 4v9 (ConvNet)	0.60%	0%	4.20%
Dogs vs Cats (ResNet50)	2.50%	0%	6.50%

The results show that WisdomNet can eliminate classification mistakes by allowing the model to reject uncertain cases instead of forcing a prediction.

FACTORS AFFECTING PERFORMANCE:

- **Base Network Accuracy:** Models with lower accuracy tend to have higher reject rates. The reject rate generally stays below 10% when the model accuracy is around 98%
- **Class Similarity:** More similar classes produce higher rejection rates. For example, similar MNIST digits result in higher reject rates than clearly distinguishable digits.
- **Noise:** The reject rate increases with noise in the data as the model's confidence decreases.

LIMITATIONS AND DISCUSSIONS:

- A single conjugate neuron does not scale well to multiclass problems, making WisdomNet effective only for binary classification. For example, when classifying five MNIST digits, it rejected more than 50% of the samples.

- The multi-conjugate-neuron approach, which uses one reject neuron per class, reduced the reject rate to 9.8% for five classes. However, performance declined as the number of classes increased.
- The method does not provide a defined criterion for determining when to stop fine-tuning. Excessive training may increase the rejection rate, while insufficient training may allow unreliable predictions to remain undetected.

CONCLUSION:

WisdomNet shows that a simple modification to an existing neural network can make its predictions more trustworthy. By adding a zero-initialized reject neuron and fine-tuning on the model's own mistakes, it achieved 0% classification error in the tested binary classification tasks while maintaining relatively low reject rates.

The approach requires no additional manual labels and can be applied to different types of neural networks. Instead of forcing the model to make a prediction when it is uncertain, WisdomNet allows it to reject the case for human review. However, efficiently extending the approach to larger multiclass problems remains an area for future research.